

Def. Given $n \in \mathbb{N}$, a polynomial

$$\Phi_n \stackrel{\text{def}}{=} \prod_{\zeta \in \mu_n, \zeta \text{ is generator of } \mu_n} (x - \zeta)$$

is called n^{th} -cyclotomic polynomial.

Note: $\mu_n = \{ \exp(2\pi i \cdot \frac{k}{n}) \mid k=0, 1, \dots, n-1 \}$ is a group, under the multiplication.

Actually, $\mu_n \cong C_n$, the cyclic group of order n .

Moreover, for all $d \mid n$, $\mu_d \leq \mu_n$. Observe that

$$x^n - 1 = \prod_{\zeta \in \mu_n} (x - \zeta) = \prod_{d \mid n} \prod_{\substack{\zeta \in \mu_n \\ \langle \zeta \rangle = \mu_d}} (x - \zeta) = \prod_{d \mid n} \Phi_d(x)$$

Thm. Let ζ_n be the n^{th} root of unity. Then,

$$(\mathbb{Z}/n\mathbb{Z})^\times \cong \text{Gal}(\mathbb{Q}(\zeta_n)/\mathbb{Q}).$$

Proof. Let

$$(\mathbb{Z}/n\mathbb{Z})^\times \rightarrow \text{Gal}(\mathbb{Q}(\zeta_n)/\mathbb{Q})$$

$$a \pmod{n} \mapsto \sigma_a \quad \text{where } \sigma_a(\zeta_n) = \zeta_n^a.$$